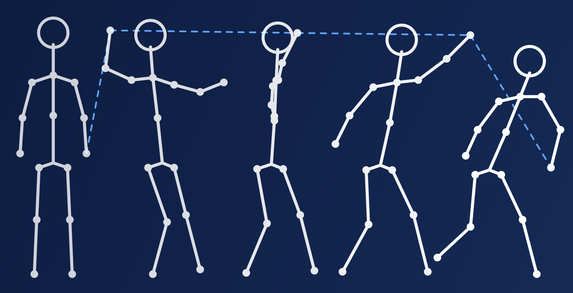
ESA-GCN

Efficient Semantic-Aware Graph Convolution for Skeleton-Based Action Recognition

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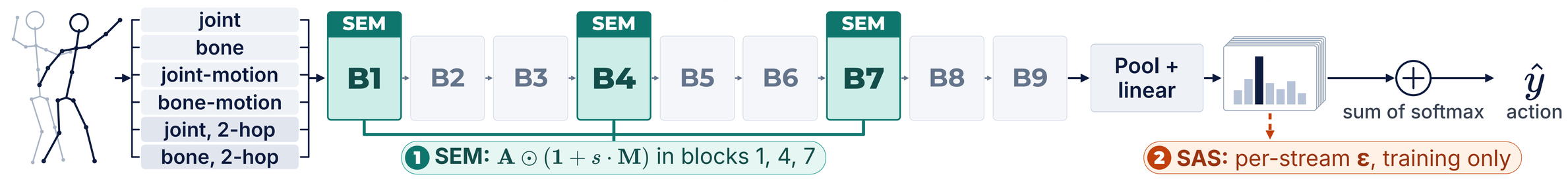


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1 SEM Semantic Edge Mask
Multiply, don't add.
A language mask multiplies the skeleton graph and avoids gate collapse.

2 SAS Stream-Asymmetric Smoothing
Learn ε, don't search it.
Each stream learns its own label smoothing: no grid search, zero inference cost.

3 ESA-GCN LA-GCN + SEM + SAS
Better on 25 / 25.
Every result beats a retrained LA-GCN, for ≤ 2.6% more parameters.

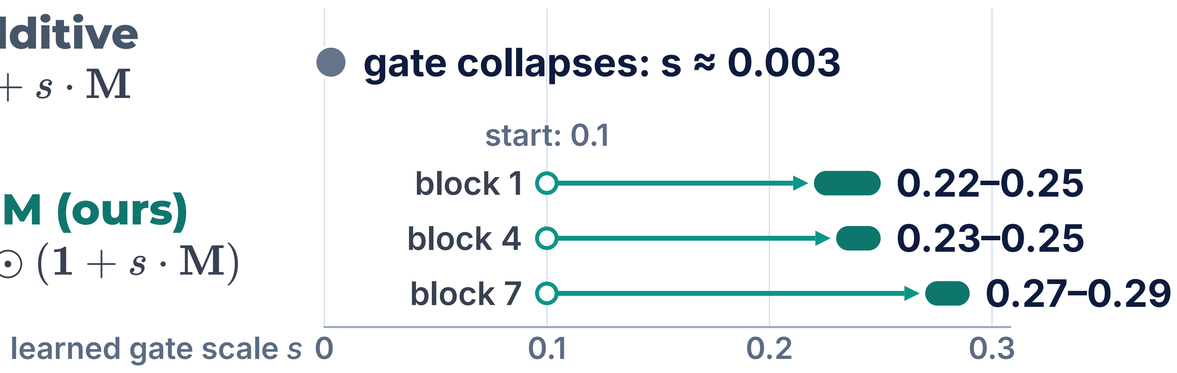


1 SEM Semantic Edge Mask

PROBLEM **Additive gates are prone to collapse**

Additive
$\mathbf{A} + s \cdot \mathbf{M}$

SEM (ours)
$\mathbf{A} \odot (\mathbf{1} + s \cdot \mathbf{M})$



SEM starts as identity, and s grows on every split.

OUR FIX **A language mask multiplies the graph**

Action classes
e.g. "drink water"
→ **Sentence-BERT**
once, offline
→ **Frozen codebook $\mathbf{E}$**
one row per class

Cross-attention
joint features $\mathbf{Q}$ attend to $\mathbf{E}$
$\mathbf{a} = \mathrm{softmax}\left(\mathbf{Q}\mathbf{E}^{\top}/\sqrt{d_e}\right)$

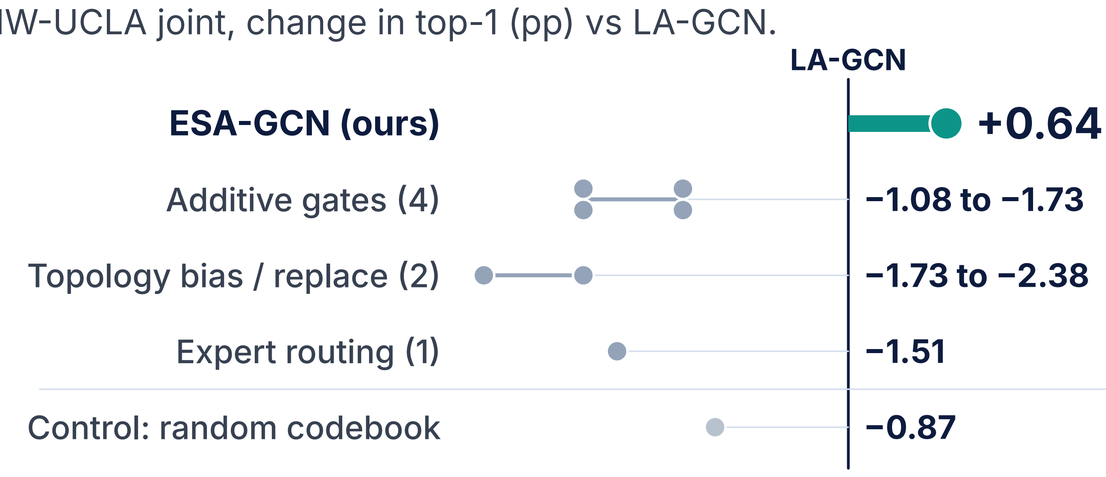
→ $\mathbf{M} = \mathbf{a}\,\mathbf{a}^{\top}$ →

Reweight graph edges
$\tilde{\mathbf{A}}_c = \mathbf{A}_c \odot (\mathbf{1} + s \cdot \mathbf{M})$

$\mathbf{M}$ links joints that attend to the same actions.

EVIDENCE **Multiply beats 7 alternatives**

NW-UCLA joint, change in top-1 (pp) vs LA-GCN.



2 SAS Stream-Asymmetric Smoothing

PROBLEM **Smoothing ε is grid-searched**

◇ **0.05** — joint, joint-motion
4×
◇ **0.20** — bone, bone-motion

Each modality needs its own ε, grid-searched per dataset.
A fourfold gap that each stream can learn instead.

OUR FIX **One learnable ε per stream**

$$\varepsilon^{(\mathrm{stream})} = \varepsilon_{\max} \cdot \sigma\left(\varepsilon_{\mathrm{logit}}^{(\mathrm{stream})}\right)$$

one learnable scalar per stream

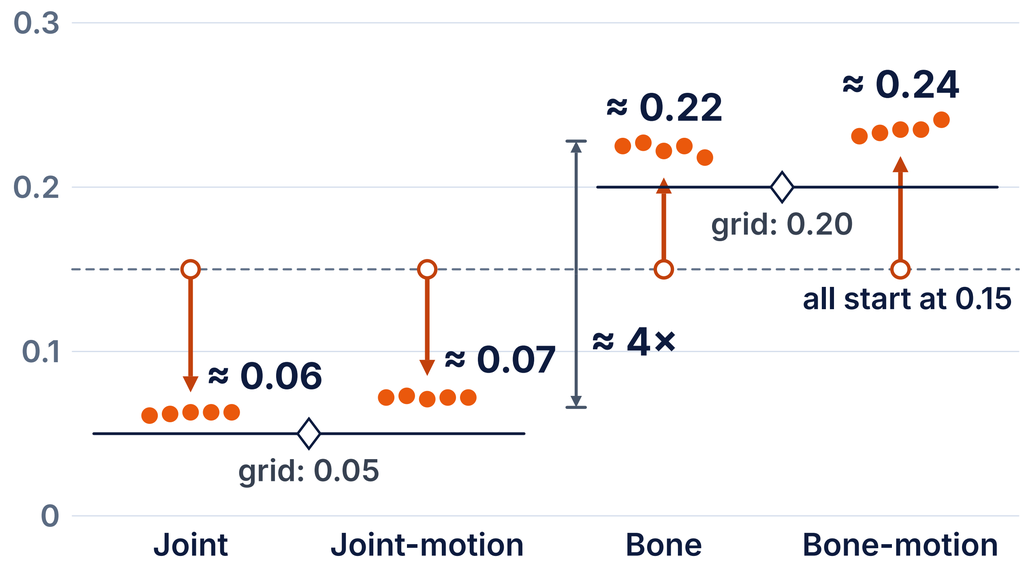
$$\tilde{y} = (1 - \varepsilon)\, y + \varepsilon / K$$

smoothed target of the training loss

- All streams start at 0.15
- $\varepsilon_{\max} = 0.3$, a soft upper bound
- **Loss only, zero inference cost**

EVIDENCE **Learned ε recovers the 4× gap**

20 trainings start at 0.15 and settle near the grid-searched values (diamonds).



3 ESA-GCN consistent gains for ≤ 2.6% more parameters and +4.9% FLOPs

Best or tied on 4 of 5 splits

Six-stream top-1 (%) among the 13 methods compared in the paper.

Method	Params (M)	NTU 60		NTU 120		NW-UCLA
		X-Sub	X-View	X-Sub	X-Set	
InfoGCN	1.57	93.0	97.1	89.8	91.2	97.0
HD-GCN	1.66	93.4	<u>97.2</u>	90.1	91.6	97.2
LA-GCN	1.44	<u>93.5</u>	<u>97.2</u>	**90.7**	<u>91.8</u>	<u>97.6</u>
ESA-GCN	1.47	**93.65**	**98.02**	<u>90.19</u>	**92.17**	**97.63†**

Comparators quoted from their papers. **Bold** best, <u>underlined</u> second. † Ties 97.6. Higher than InfoGCN and HD-GCN on every split, with fewer parameters per stream.

25 / 25 beat retrained LA-GCN

	Joint	Bone	J-motion	B-motion	4-stream
60 X-Sub	+0.35	+0.81	+0.50	+1.34	+0.53
60 X-View	+0.53	+0.81	+0.50	+1.16	+0.53
120 X-Sub	+0.64	+0.81	+0.66	+1.37	+0.73
120 X-Set	+0.63	+0.77	+0.70	+1.37	+0.73
NW-UCLA	+0.64	+1.94	+0.44	+2.15	+0.64

Top-1 gain (pp) over LA-GCN retrained in our pipeline and schedule. 4-stream gain +0.53 to +0.73 on every split.

Take-home

Inject language by multiplying the skeleton graph, not adding to it, and let each stream learn its own label smoothing.
Two compact modules beat a retrained LA-GCN on all 25 results for ≤ 2.6% more parameters.

Xu et al., LA-GCN, IEEE TMM 2025. Chen et al., CTR-GCN, ICCV 2021. Reimers and Gurevych, Sentence-BERT, EMNLP 2019.

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